

pdac-recurrent-cldn18-2-kras-g12r-tp53 -y220c-p25m

Preclinical recommendations — forward-looking horizon scan

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Preclinical recommendations

A forward-looking horizon scan of candidate drugs, compounds, and treatment strategies that are earlier than clinical development, or not yet developed, but plausibly target this patient's stated features. These are research directions, not enrollable options or treatment recommendations.

1. [177Lu]Lu-DOTA-zolbetuximab (CLDN18.2 antibody radioconjugate) (radioligand)

Target: CLDN18.2 expression, 2+/3+ in 60% of tumor cells **Stage:** In vivo (animal) **Evidence strength:** Moderate

Best mechanistic match in the case and currently unreachable: the crossfire argument fits a patchy 60% antigen, but every open trial in the class is closed to this patient.

Of everything in this file, this is the only candidate whose supporting in-vivo work was done in PDAC xenografts, and the only mechanism that makes a patchy antigen survivable. Beta emission from lutetium-177 deposits dose across roughly a millimetre, so the 40% of cells that stain negative sit inside the kill radius of the 60% that do not. That matters here, because 2+/3+ in 60% of cells is a positive stain but a below-cutoff one. It falls short of the 75%-of-cells bar used to register zolbetuximab, and a sub-cutoff result predicts benefit less reliably than a high-positive one, so the antigen argument should be read as real but soft. The stain is also from the December 2025 resection, not from the recurrence. One thing has changed since the scan was written: the class crossed into humans. Two CLDN18.2 radioligand studies are recruiting (NCT07595237, NCT07707531), and NCT07595237's threshold of 2+ staining in at least 20% of cells is cleared roughly threefold by this tumor. The accessibility screen still graded both not_yet_accessible, because they are single-centre Chinese investigator-initiated studies with no route for a foreign enrollee, and a worldwide sweep turned up no Western CLDN18.2 radioligand program at any stage. So this candidate sits in an odd place: the best mechanistic match in the case, and out of reach at the same time.

Mechanism. A CLDN18.2 antibody carries lutetium-177 to the tumor, and the beta emission deposits dose across a path of roughly a millimetre. Killing therefore does not need antigen on every cell: cells within crossfire range of a bound antibody take the dose too. A paired 89Zr imaging agent measures target engagement before treatment instead of inferring it from an archival stain.

Translatability. High for a horizon item: the only candidate with in-vivo data in PDAC xenografts, though a 60% antigen fraction is thinner than those lines carried.

Developability. Short path by horizon standards: approved antibody, routine lutetium-177 chemistry. Needs GMP radiolabelling, dosimetry, and marrow and renal safety before first-in-human.

Counter-productive MoA. Low (Crossfire kills antigen-negative cells too, so heterogeneity costs dose rather than working against the mechanism)

Key risks.

- CLDN18.2 sits on normal gastric mucosa and a radioconjugate irradiates it; human gastric and marrow toxicity is unmeasured
- Subcutaneous xenografts lack the desmoplastic barrier that limits antibody delivery in real PDAC, so uptake in

this tumor would likely be lower

- The 60% antigen fraction is below what the xenograft lines expressed and below the zolbetuximab registrational cutoff; crossfire softens that but does not erase it
- The 60% call comes from the resected primary; CLDN18.2 on the recurrent lesion has never been measured
- One publication from one group, with no independent replication
- Both open trials in this class are single-centre Chinese investigator-initiated studies with no enrollment mechanism for this patient, and no Western program exists at any stage
- Lutetium-177 supply and radiopharmacy capacity limit where this could ever be delivered

Open questions.

- Does the recurrent lesion still carry CLDN18.2, and at what fraction of cells?
- Would an 89Zr companion scan show enough uptake through PDAC stroma to justify a therapy dose?
- The flip event on access is a US IND from SmartNuclide or any Western entrant into CLDN18.2 radioligands. Nothing at present suggests a date.

Key references. pmid:41266253

2. ACBI4 (VHL-recruiting PROTAC degrader of GTP-loaded KRAS) (PROTAC / degrader)

Target: KRAS p.G12R **Stage:** In vitro **Evidence strength:** Emerging

The most direct answer to the G12R chemistry vacuum, resting on a single in-vitro paper with no animal data and no pancreatic model behind it.

G12R has no selective inhibitor, and the covalent arginine chemistry that once looked like the way in has not moved past a crystal structure in four years. ACBI4 sidesteps that problem by not needing allele-specific chemistry at all: it recruits VHL to KRAS in its GTP-loaded state and degrades it, so an allele with no covalent handle becomes addressable anyway. Degradation also strips the scaffolding role of KRAS, which occupancy inhibitors leave alone, and it would take out the wild-type RAS-GTP that G12R cells reportedly lean on when they escape daraxonasib. What holds it at second rather than first is how thin the file is. One paper, one group, cell lines only, no pancreatic model, nothing dosed in an animal.

Mechanism. ACBI4 bridges the VHL E3 ligase to KRAS in its GTP-loaded state and routes the protein to the proteasome. The handle is the active conformation rather than a mutant-specific cysteine or aspartate, so alleles with no covalent chemistry of their own become addressable. Degradation also removes the scaffolding functions of KRAS, which occupancy inhibitors leave intact.

Translatability. Moderate: potent G12R degradation in KRAS-mutant cell lines, but no pancreatic model and nothing dosed in an animal tumor yet.

Developability. Tool compound. Bifunctional VHL degraders this size have permeability and oral-exposure problems; a drug-like molecule, PDAC in-vivo work, and an IND are all missing.

Counter-productive MoA. Moderate (Degrading GTP-loaded KRAS also strips wild-type RAS-GTP from normal cells; the therapeutic index is unmeasured)

Key risks.

- No in-vivo efficacy or tolerability data; potency in cells says little about exposure in tissue
- Degrading GTP-loaded KRAS also hits wild-type RAS-GTP in normal cells, and the therapeutic index has not been measured
- The paper contains no PDAC model, so the G12R activity was shown in lines of other lineages
- VHL-based degraders are vulnerable to ligase downregulation as a resistance route
- One paper, no independent replication of the G12R result

Open questions.

- Does degradation of GTP-loaded KRAS hold up in a G12R pancreatic model in vivo?
- Is there a window between tumor KRAS and wild-type RAS-GTP in normal tissue?
- Would VHL downregulation close this route the way it has for other VHL degraders?

Key references. pmid:41166656

3. CDK8/19 inhibition added to RAS(ON) inhibition (with or without CTLA-4 blockade) (combination)

Target: KRAS p.G12R **Stage:** In vivo (animal) **Evidence strength:** Moderate

In-vivo PDAC data on a resistance mechanism the patient's most likely drug class would actually hit, built on one lab's single paper with a G12D-selective tool.

This tumor is MSS with a TMB of 0.83, so any immune route has to be manufactured rather than found, and RAS inhibition is what manufactures it. Kalluri's group showed CDK8 closing the immune window that KRAS blockade opens, through CXCL2 induction and FAS suppression, and showed that hitting CDK8 alone or with anti-CTLA-4 reversed resistance in vivo. The detail that makes it relevant here is that CDK8 was also up in PDX tumors resistant to daraxonrasib and to zoldonrasib, which is the class this patient would most plausibly see. The inference is not clean. The tool compound was G12D-selective, and G12R escapes differently, leaning on EGFR-driven wild-type RAS rather than the mutant allele, so whether the same CDK8 program governs escape in this tumor is untested. One lab, one publication, no independent replication yet.

Mechanism. Blocking mutant KRAS in PDAC opens a short immune window, with dendritic cells and T cells moving in next to cancer cells. CDK8, a mediator-complex kinase, closes it again by driving CXCL2 secretion and holding FAS expression down, and the microenvironment reverts to an immune-evasive state that carries the tumor into resistance. Inhibiting CDK8 during RAS suppression holds the window open instead of trying to reopen it later.

Translatability. Moderate: in-vivo PDAC data, but generated with a G12D-selective compound, so the extension to a G12R tumor is an inference.

Developability. Clinical-grade CDK8/19 inhibitors exist in other indications, so the gap is a combination protocol and tolerability data on top of a pan-RAS backbone.

Counter-productive MoA. Moderate (CDK8 loss is tumor-suppressive in some contexts, so the direction of effect is not guaranteed)

Key risks.

- The in-vivo work used a G12D-selective tool compound; whether CDK8 governs escape in a G12R tumor is untested, and G12R adapts through EGFR-driven wild-type RAS
- CDK8 loss is tumor-suppressive in some contexts and tumor-promoting in others, so the direction of effect is not guaranteed
- Adding CTLA-4 blockade brings real human toxicity for a benefit shown only in mice
- Overlapping myelosuppression and GI toxicity with a RAS backbone has not been quantified
- Single laboratory, single publication

Open questions.

- Is CDK8 upregulated in G12R tumors escaping a pan-RAS inhibitor, or only in G12D ones?
- Can a CDK8/19 inhibitor be added to a pan-RAS backbone at doses either drug needs?
- Does the CTLA-4 arm add anything once CDK8 is blocked, or just toxicity?

Key references. pmid:42436354

4. p53 Y220C reactivation paired with a BH3 mimetic to raise apoptotic priming (combination)

Target: TP53 p.Y220C **Stage:** In vivo (animal) **Evidence strength:** Emerging

An apoptotic-priming rescue nobody has built, with both components clinical-grade; the supporting biology is genetic p53 restoration in mouse lung, not Y220C in pancreas.

Whether restored p53 kills a cell or merely stops it dividing is decided before p53 gets there, by how close the mitochondria already sit to the apoptotic threshold. KRAS-driven epithelial tumors sit far from it and tend to arrest. Sanchez-Rivera's group showed exactly that in Kras-driven mouse tumors, then showed navitoclax or BCL-2/BCL-XL deletion converting arrest into apoptosis, in vitro and in vivo. This case carries both halves of that configuration, a Y220C reactivator route and a KRAS driver, and nobody has built the combination. If rezatapopt-class agents underperform in KRAS-mutant solid tumors, low priming is a specific and testable reason rather than a shrug, and BH3 profiling could say in advance whether it applies. The gap is that the supporting work restored wild-type p53 genetically, in mouse lung and sarcoma, not pharmacologically in a Y220C pancreatic tumor.

Mechanism. Whether restored p53 kills a cell or only stops it dividing is set upstream of p53, by how close the mitochondria already sit to the apoptotic threshold. KRAS-driven epithelial tumors sit far from it, holding reserve BCL-2 and BCL-XL capacity, and they arrest. A BH3 mimetic occupies that reserve, so the transcriptional output of restored p53 lands on a primed mitochondrion and the same signal produces apoptosis instead.

Translatability. Moderate: convincing in-vivo priming biology, but in mouse lung and sarcoma with genetic p53 restoration, not pharmacologic Y220C rescue in pancreas.

Developability. Both halves already exist as clinical-grade molecules, so this is combination design rather than discovery. Missing: the experiment itself, in a KRAS-mutant Y220C PDAC model.

Counter-productive MoA. Moderate (Navitoclax works through BCL-XL, the same arm that causes dose-limiting thrombocytopenia, capping the priming actually achievable)

Key risks.

- Navitoclax brings dose-limiting thrombocytopenia through BCL-XL, the arm doing the work here; a BCL-XL degrader would route around it but adds its own unknowns
- The supporting model restored wild-type p53 genetically, a cleaner and probably stronger signal than a small molecule refolding Y220C
- No pancreatic model and no Y220C model anywhere in the supporting work
- Marrow toxicity would overlap with anything cytotoxic this patient might also need

Open questions.

- Would BH3 profiling of this tumor show the low priming the hypothesis assumes?
- Does pharmacologic Y220C refolding produce the same transcriptional output that genetic restoration did?
- Can a BCL-XL degrader deliver the priming without navitoclax's thrombocytopenia?

Key references. pmid:34074758

5. Statin-driven degradation of misfolded p53 via the mevalonate-DNAJA1 axis (repurposed drug)

Target: TP53 p.Y220C **Stage:** In vitro **Evidence strength:** Emerging

Attacks the same lesion as a reactivator from the opposite direction and therefore rules it out; cheap to test, but the evidence stops at 2016 cell culture.

Iwakuma's screen found that statins deplete conformational p53 mutants specifically, and the Y220C example was BxPC-3, a pancreatic line. For a chemical-biology paper that is an unusually close match to this patient's variant and tumor type, which is most of why it ranks where it does. Statins are off-patent, oral, and familiar. The problem is that this route and a Y220C reactivator are mutually exclusive rather than complementary. Rezatapoct-class drugs need the misfolded protein present in order to refold it, and this approach destroys it, so choosing one closes the other. That is a decision to make before either is started, not something to discover afterwards. The evidence is also a decade old, in vitro, with little independent replication in pancreatic models, and in-vitro statin concentrations routinely exceed what oral dosing reaches in tissue.

Mechanism. Y220C destabilises the DNA-binding domain, and the misfolded protein survives because DNAJA1, an Hsp40 chaperone, holds it away from CHIP-mediated ubiquitylation. Statins cut mevalonate-5-phosphate, which weakens the DNAJA1 interaction and releases the mutant protein for nuclear export and proteasomal degradation. The approach removes the mutant protein rather than refolding it.

Translatability. Moderate: the one paper used a pancreatic Y220C line, BxPC-3, which is close to this case, but the work stops at cell culture.

Developability. Closest to hand of anything here: off-patent oral drug, investigator-initiated window study feasible. Whether achievable plasma levels deplete mutant p53 is unanswered.

Counter-productive MoA. High (Degrading mutant p53 removes the substrate a Y220C reactivator needs; the two routes cannot both be used)

Key risks.

- Directly antagonistic to the rezatapoct-class reactivator route this case otherwise has: degradation removes the

protein a reactivator is meant to refold, so the two cannot be combined and cannot be casually sequenced

- In-vitro statin concentrations often exceed what oral dosing achieves in tumor tissue, the usual failure point for statin repurposing
- The benefit assumes mutant p53 is doing gain-of-function work in this tumor, which has not been shown for this patient
- Retrospective statin-and-cancer epidemiology is confounded and is not support for this mechanism
- A decade-old finding with limited replication in pancreatic models and no in-vivo Y220C experiment

Open questions.

- Does an achievable oral statin dose lower mutant p53 protein in a Y220C pancreatic model?
- Is mutant p53 in this tumor doing anything that removing it would help?
- Which route does the case want, refolding or degradation? Both cannot be pursued.

Key references. pmid:27775703

6. Purine-synthesis blockade in COMPASS-mutant cells (lometrexol / GARFT inhibition) (repurposed drug)

Target: KMT2C p.S2816fs **Stage:** In vivo (animal) **Evidence strength:** Emerging

The concrete half of an otherwise untranslatable variant: a named metabolic dependency with animal data, separated from this patient by a mouse ES-cell double knockout.

The standing verdict on KMT2C S2816fs in this case was that nothing translatable exists. That is too strong. Shilatifard's group ran CRISPR dropout screens in MLL3/4-depleted cells, found synthetic lethality on purine and pyrimidine synthesis, and showed MLL3/4-mutant tumors were sensitive to lometrexol in culture and in animals. A named dependency with an in-vivo readout beats a general 'try an epigenetic agent' hypothesis. The catch is distance. The supporting biology is a double MLL3/MLL4 knockout in mouse embryonic stem cells, while this tumor has a single KMT2C frameshift, reported heterozygous, in human pancreas. Whether S2816fs abolishes the enhancer function that drives the metabolic shift has never been tested for this variant, and there is no pancreatic model anywhere in the work.

Mechanism. Losing MLL3/MLL4 enhancer function shifts metabolism toward de novo purine synthesis, with MLL1/COMPASS compensation underlying the shift. The cell then leans on a pathway the intact cell does not need as badly. Lometrexol inhibits GAR formyltransferase in that pathway, which turns the dependency into a drug target.

Translatability. Low, cross-tumor only: mouse ES-cell double knockout of MLL3/MLL4 versus one heterozygous KMT2C frameshift in human pancreas, with no PDAC model anywhere.

Developability. Lometrexol was dosed in humans and shelved for toxicity: pharmacology exists, sponsor does not. A modern attempt would use another antifolate with folate rescue.

Counter-productive MoA. Low (Purine blockade adds marrow and mucosal toxicity in a heavily pretreated patient without opposing the therapeutic goal)

Key risks.

- Lometrexol's clinical history is severe and unpredictable myelosuppression and mucositis, which is why it was abandoned
- The supporting biology is a double MLL3/MLL4 knockout in mouse ES cells; a single heterozygous KMT2C frameshift in a human tumor may not create a dependency of the same depth
- No pancreatic model anywhere in the supporting work
- Whether S2816fs actually abolishes the relevant enhancer function has not been shown for this variant

Open questions.

- Does a single heterozygous KMT2C frameshift shift purine metabolism the way a double MLL3/MLL4 knockout does?
- Is there a pancreatic model carrying a KMT2C frameshift that could test this at all?
- Could a newer purine-pathway agent reproduce the effect without lometrexol's toxicity history?

Key references. pmid:37252797

7. CLDN18.2 CAR-T armored with secreted hyaluronidase (sPH20-IgG2) (cell therapy)

Target: CLDN18.2 expression, 2+/3+ in 60% of tumor cells **Stage:** In vivo (animal) **Evidence strength:** Emerging

A construct-level fix for the stromal exclusion that has blunted CLDN18.2 cell therapy in pancreas, inferred entirely from CARs built against other antigens.

The usual explanation for CLDN18.2 cell therapy reading better in gastric cancer than in pancreas is stroma, not antigen, and the satri-cel experience in this case's files is mostly gastric. Zhao's group showed hyaluronan physically restricting CAR-T mobility, a secreted PH20-IgG2 module restoring transmigration without blunting cytotoxicity, and armored cells producing smaller tumors in vivo. Adding that module to a CLDN18.2 binder is a construct change, not a new target. The caveat is large and should not be smoothed over: every supporting experiment used mesothelin or GPC3, in gastric and liver models. No CLDN18.2 version has ever been built, so this is class-level inference. The antigen picture is soft too. 60% of cells at 2+/3+ from archival primary tissue is below the zolbetuximab cutoff and unverified on the recurrence, and a low-positive stain predicts benefit less reliably than a high-positive one. Unlike a radioconjugate, a CAR needs antigen on the cell it is killing, so heterogeneity costs more here than it does at rank 1.

Mechanism. Hyaluronan-rich matrix physically restricts T-cell mobility in PDAC, so transferred cells stall before reaching antigen. Engineering the CAR construct to secrete PH20 hyaluronidase puts the enzyme where the cells are, continuously, rather than relying on a systemic dose of the kind that failed in HALO-301. The armoring is a delivery fix and leaves the CLDN18.2 recognition domain untouched.

Translatability. Low, cross-tumor only: hyaluronidase armoring was shown with mesothelin and GPC3 CARs in gastric and liver models, never with a CLDN18.2 binder.

Developability. Routine cell engineering, but it means a new IND-stage product rather than a tweak to an existing one, and manufacturing time fights progressive disease.

Counter-productive MoA. Moderate (Degrading matrix to let T cells in could also ease tumor-cell dissemination out)

Key risks.

- Every supporting experiment used a different antigen, mesothelin or GPC3, in non-pancreatic models; no CLDN18.2 version has been built
- A CAR needs antigen on the cell it kills, and 40% of cells here stained negative on archival primary tissue, below the zolbetuximab cutoff and unverified on the recurrence
- On-target gastric mucosal toxicity of CLDN18.2 CAR-T is construct-independent, and better tumor penetration does nothing to reduce it
- Systemic hyaluronidase caused thromboembolic events in the PEGPH20 program; local secretion should lower but not remove that concern

Open questions.

- Does PH20 armoring survive the move to a CLDN18.2 binder, or was the effect antigen-dependent?
- What antigen fraction does a CLDN18.2 CAR need, and does 60% on archival tissue clear it?
- Is local PH20 secretion safe enough given what systemic PEGPH20 did?

Key references. pmid:34326835; pmid:33347692

8. PARP inhibition gated on KMT2C loss rather than BRCA status (repurposed drug)

Target: KMT2C p.S2816fs **Stage:** In vivo (animal) **Evidence strength:** Emerging

A claim about which biomarker opens the PARP door, not about a new drug, and this patient's platinum history argues against the repair defect it assumes.

Chang and colleagues found KMT2C at DNA damage sites doing something unrelated to promoter transcription, recruited by Ago2 and damage-response RNAs, and showed KMT2C/D mutations disrupting homologous recombination and sensitising NSCLC models to PARP inhibitors. They proposed the mutation as a biomarker where BRCA1/2 mutations are rare. This patient has a KMT2C frameshift and a pending germline panel, so if the germline comes back clean this is the only remaining argument for a PARP inhibitor in the case. It is also an argument the clinical course pushes back on. Ten cycles of platinum-containing therapy without durable control is not what a deep HR defect usually looks like, and the frameshift is reported heterozygous, which may leave enough function for repair to run. One lab, NSCLC only, thin replication elsewhere. None of this is a reason to give a PARP inhibitor today.

Mechanism. KMT2C has a role at DNA damage sites unrelated to promoter transcription: Ago2 and damage-response RNAs recruit it, it methylates H3K4 locally, chromatin relaxes, and repair factors assemble. Losing it degrades homologous recombination, the same functional state that makes BRCA-mutant tumors sensitive to PARP inhibitors. The gate here is a chromatin gene rather than a canonical HR gene.

Translatability. Low, cross-tumor only: one lab's NSCLC result, and this tumor's behaviour across ten platinum-containing cycles argues against a deep repair defect.

Developability. Drugs are approved, so the work is biomarker qualification. The gating test is a functional HRD readout, and none is standardised for pancreas.

Counter-productive MoA. Low (PARP inhibition without a real repair defect costs marrow reserve and buys

nothing)

Key risks.

- A single laboratory's result in NSCLC, with thin independent replication in other tumor types
- KMT2C S2816fs is heterozygous on the report and may leave enough function for HR to stay intact
- Platinum exposure across 10 FOLFIRINOX cycles without durable control argues against a deep HR defect in this tumor
- Functional HRD assays for pancreatic cancer are not standardised, so the gating test does not really exist yet

Open questions.

- Does a functional HRD assay on this tumor show any defect, given ten cycles of platinum without durable control?
- Is a heterozygous KMT2C frameshift enough to disrupt repair, or does the NSCLC result depend on deeper loss?
- What does the pending germline panel show, since a BRCA-family finding would settle the question a different way?

Key references. pmid:33853832

9. Pharmacologic induction of CLDN18.2 (hypomethylating agent as antigen-density priming) (repurposed drug)

Target: CLDN18.2 expression, 2+/3+ in 60% of tumor cells **Stage:** In vitro **Evidence strength:** Speculative

Aims directly at the 40% of cells that stain negative, on the strength of one 2011 cell-line paper where demethylation was an amplifier, not a driver.

The weakest point in this patient's strongest handle is arithmetic. 60% of cells carry the antigen and 40% do not, which is what puts the tumor under the zolbetuximab registrational cutoff and makes the stain a softer predictor of benefit than a high-positive result would be. Raising antigen on the negative fraction would address the eligibility problem head-on instead of routing around it, and Ito's group showed 5-aza-deoxycytidine amplifying CLDN18a2 induction in HPAF-II and HPAC. The evidence needs reading carefully, though. That was 2011, in cell lines, and demethylation acted as an amplifier of phorbol-ester-driven induction rather than as a standalone inducer, so the standalone effect may be small or absent. Nobody has shown that a tolerable human dose moves CLDN18.2 protein at the membrane, which is the only thing an IHC cutoff or a CAR responds to.

Mechanism. CLDN18.2 transcription in pancreatic cells runs through PKC and MAPK signaling onto an AP-1 site, and a CpG island inside the CLDN18 coding sequence gates how much of that signal gets through. Demethylation lifts the gate. Raising antigen on the cells that currently stain negative would matter more for a threshold-gated therapy than any change in the cells already staining 3+.

Translatability. Low: 2011 cell-line work where demethylation amplified phorbol-ester-driven induction rather than acting alone, with no in-vivo or patient-tissue confirmation.

Developability. Decitabine and azacitidine are approved and cheap, so a priming schedule is easy to write. Nobody has shown a tolerable dose moves membrane protein.

Counter-productive MoA. Moderate (Raising CLDN18.2 on normal gastric mucosa would widen the on-target toxicity margin for any CLDN18.2 therapy)

Key risks.

- The effect was an amplifier of phorbol-ester-driven induction rather than standalone demethylation, so the standalone effect may be small or absent
- Raising CLDN18.2 on normal gastric mucosa would widen the on-target toxicity problem for any CLDN18.2 therapy
- Hypomethylating agents cause myelosuppression, a real cost in a patient heading toward cell therapy or more chemotherapy
- Fifteen-year-old cell-line data with no in-vivo or patient-tissue confirmation

Open questions.

- Does a tolerable decitabine or azacitidine schedule move CLDN18.2 protein at the membrane, not just mRNA?
- Would induced expression be uniform enough to change an IHC percentage, or patchy?
- How much does the same schedule raise CLDN18.2 on normal gastric mucosa?

Key references. pmid:21381080

10. PI3Kgamma blockade of KRAS-independent macropinocytosis (eganelisib / IPI-549 as the available tool) (repurposed drug)

Target: KRAS p.G12R **Stage:** In vitro **Evidence strength:** Speculative

The weakest evidence in this scan: one macropinocytosis assay, no growth or animal readout, and an allele-selectivity claim the same paper partly undercuts.

This is the thinnest evidence in the file, and it ranks accordingly despite an attractive story. The story: G12R cannot engage p110alpha because switch II is distorted, so it runs macropinocytic scavenging through KRAS-independent PI3Kgamma instead, a supply line a RAS inhibitor would not close. The evidence: one paper, one assay, 300 nM IPI-549 lowering macropinocytosis in G12R lines, with no growth endpoint, no survival endpoint, and no animal data. Worse for the framing, IPI-549 also cut macropinocytosis in G12D and G12V lines, so the allele-selectivity premise is shakier than the mechanism narrative makes it sound. Eganelisib has human pharmacology from unrelated myeloid work, which is the only reason this would be easy to test. Somebody would still have to show that blocking scavenging slows a G12R pancreatic model before it means anything.

Mechanism. G12R cannot engage p110alpha because switch II is distorted, so it does not drive macropinocytosis the way G12D and G12V do. Hobbs and colleagues found that G12R PDAC lines instead run macropinocytic nutrient scavenging through KRAS-independent PI3Kgamma and RAC1. Blocking p110gamma would cut a supply line that in this allele sits outside the KRAS node, which is also why a RAS inhibitor would not close it.

Translatability. Low: a single uptake assay in G12R lines with no growth endpoint, and the same drug cut macropinocytosis in G12D and G12V too.

Developability. Eganelisib has been through human dosing, so a window study is writable. Nothing yet shows that blocking scavenging slows a G12R tumor.

Counter-productive MoA. Moderate (PI3Kgamma blockade reshapes myeloid cells, which could cut either way in a T-cell-poor pancreatic microenvironment)

Key risks.

- IPI-549 also reduced macropinocytosis in G12D and G12V lines, so the effect is not G12R-selective and the allele rationale is weaker than the framing suggests
- No evidence that blocking scavenging translates into growth inhibition, let alone regression
- PI3Kgamma inhibition reshapes myeloid cells, which could cut either way in a T-cell-poor pancreatic microenvironment
- Nutrient scavenging has redundant routes, autophagy and entosis among them, that would likely compensate
- One paper, one assay endpoint, no replication and no animal data

Open questions.

- Does blocking macropinocytosis slow a G12R pancreatic model, or only lower dye uptake?
- Is any part of the effect actually G12R-specific, given what happened in G12D and G12V lines?
- Would autophagy pick up the slack the moment scavenging is blocked?

Key references. pmid:31649109

11. KRAS G12R-directed TCR discovery and ex-vivo priming (allele extension of an existing platform) (conceptual strategy)

Target: KRAS p.G12R **Stage:** Conceptual **Evidence strength:** Speculative

A discovery step nobody has taken rather than a candidate therapy, gated on whether this patient's HLA presents the G12R peptide at all.

Every mutant-KRAS cell therapy this patient could be offered is restricted to G12D or G12V, and the upstream files record that as an eligibility exclusion rather than a biological one. In a tumor that is MSS with a TMB of 0.83, a shared driver neoantigen is one of the few antigens certainly present. Conn's group ran a single ex-vivo priming process against 46 of 47 evaluated KRAS neoantigens across 20 healthy donors, cloned more than 150 receptors, and named G12R as a target the process could be applied to but was not. The gap is a discovery step nobody has taken, not a biological wall. It also means nothing exists to give a patient. This patient's HLA type is not in the record, and whether the G12R peptide is presented on it is an open empirical question.

Mechanism. Mutant KRAS peptides are shared, public neoantigens, and TCR-T against G12D and G12V works by redirecting T cells to those peptides on the patient's HLA. A published ex-vivo priming process starts from healthy-donor blood, primes against a chosen KRAS neoepitope on the donor's HLA background, and clones the resulting receptors. Running it against the G12R peptide would produce a reagent that does not currently exist.

Translatability. Weak case match: the platform primed against 46 KRAS neoantigens but never G12R, and this patient's HLA type is not in the record.

Developability. Nothing exists yet. Epitope confirmation, receptor isolation, cross-reactivity screening, manufacturing and an IND is several years even if the biology cooperates.

Counter-productive MoA. Moderate (Engineered receptors carry alloreactivity and cross-reactivity risk; fatal

off-target events have occurred with other TCR products)

Key risks.

- The G12R peptide may simply not be presented on this patient's HLA type, and nobody has answered that empirically
- HLA typing for this case is not in the record, so eligibility for any such product is unknown
- Off-target and alloreactivity screening for a new receptor is a real safety burden; fatal cross-reactivity has happened with other engineered TCRs
- Pancreatic stromal exclusion has blunted every adoptive-cell approach tried so far, whatever the antigen

Open questions.

- Is the G12R neoepitope presented on common HLA alleles, and on this patient's in particular?
- What is this patient's HLA type? It is not in the record.
- Would a G12R TCR product get past pancreatic stromal exclusion when other adoptive approaches have not?

Key references. pmid:40359936

12. Targeting the neural-like progenitor regeneration state that survives chemotherapy (GLIS3, IL-1beta axis) (conceptual strategy)

Target: Clinical course: node-negative R0 resection with early recurrence **Stage:** Conceptual **Evidence strength:** Speculative

A research question rather than a candidate: it explains this patient's recurrence pattern better than anything else here, and offers nothing to act on.

A good pathological response, 0 of 34 nodes, regression grade 2, and then a new mass four months after adjuvant therapy ended. That shape looks more like residual disease sitting in a persistent state than like bulk chemoresistance, and Gong's work gives the observation a mechanism. The neural-like progenitor program enriched in post-treatment residual PDAC matches the normal pancreatic injury-repair response, with GLIS3 driving clonogenicity and metastasis in isogenic mouse organoids and IL-1beta signaling implicated as a microenvironmental driver. It ranks last because there is nothing to give anyone. GLIS3 is an undrugged transcription factor with no tool compound, the IL-1beta link is associative, IL-1 blockade in pancreatic cancer has already disappointed, and a program shared with normal tissue repair is a poor bet for a therapeutic window. The recurrence has not been biopsied, so whether the state is present here is currently unknowable. It stays in the file so the early-recurrence feature does not quietly disappear.

Mechanism. Residual PDAC after neoadjuvant treatment is enriched for a neural-like progenitor transcriptional program, which recent work reframes as the normal pancreatic injury-regeneration response co-opted by tumor cells recovering from cytotoxic damage. GLIS3 came out of that work as the transcription factor driving clonogenicity, growth, and metastasis in the state, with IL-1beta signaling implicated from the microenvironment side. Suppressing the state rather than the bulk tumor is the idea.

Translatability. Low: descriptive human residual-disease transcriptomes plus mouse organoid overexpression, with no compound and no target validation behind the nominated node.

Developability. No tool compound for GLIS3, an undrugged transcription factor. The nearest tractable edge is

IL-1beta blockade, which has already disappointed in pancreatic cancer.

Counter-productive MoA. Moderate (The program is shared with normal pancreatic injury repair, so suppressing it may not spare healthy tissue)

Key risks.

- No compound, no target validation, and a transcription factor as the nominated node
- A regeneration program shared with normal pancreatic injury repair is a poor candidate for a wide therapeutic window
- The IL-1beta link is associative and the causal experiments have not been done
- This patient's recurrence has not been biopsied, so whether the state is present is unknown and currently unknowable

Open questions.

- Is the neural-like progenitor state actually present in this patient's recurrence? A biopsy would be the only way to know.
- Is GLIS3 druggable by any indirect route, given it is a transcription factor?
- Is the IL-1beta association causal, and would blockade do anything after prior IL-1 failures in pancreatic cancer?

Key references. pmid:42048185